

Towards fast and constant-time implementations of Hamming Quasi-Cyclic

A case study on ARM Cortex-M4 devices

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Overview

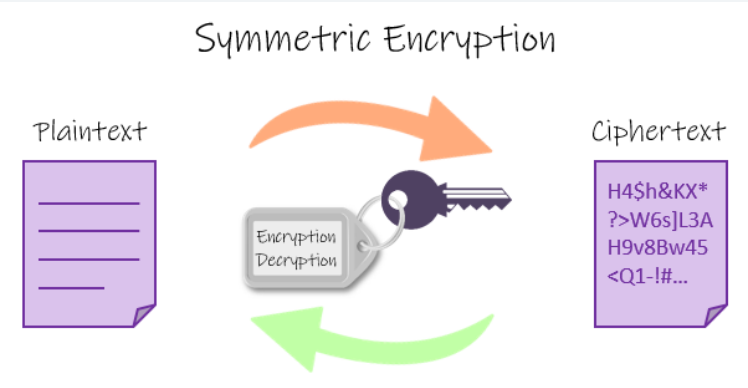
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Introduction

Kerchoff's Principle

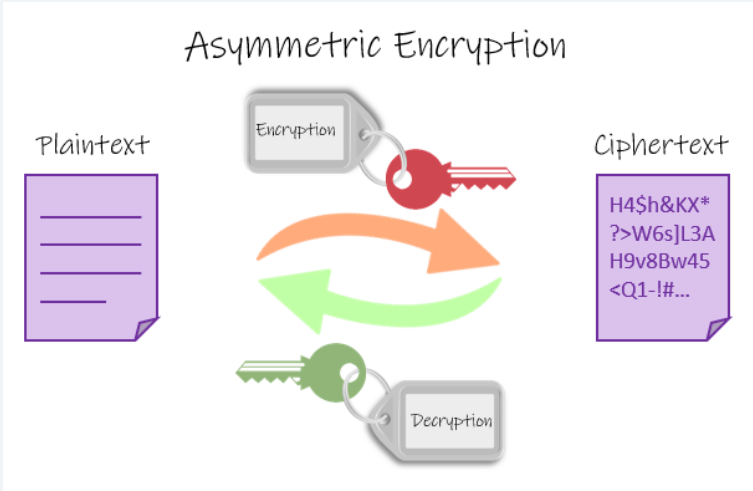
Everything about a cryptosystem is publicly known, **except for the key**

Symmetric encryption



Key: a n-bit string

Asymmetric encryption



Key: the product of two prime numbers, *i.e.* $n = p \cdot q$

Cryptanalytic attacks

What are the best attacks on modern cryptosystems?

On Classical Computers

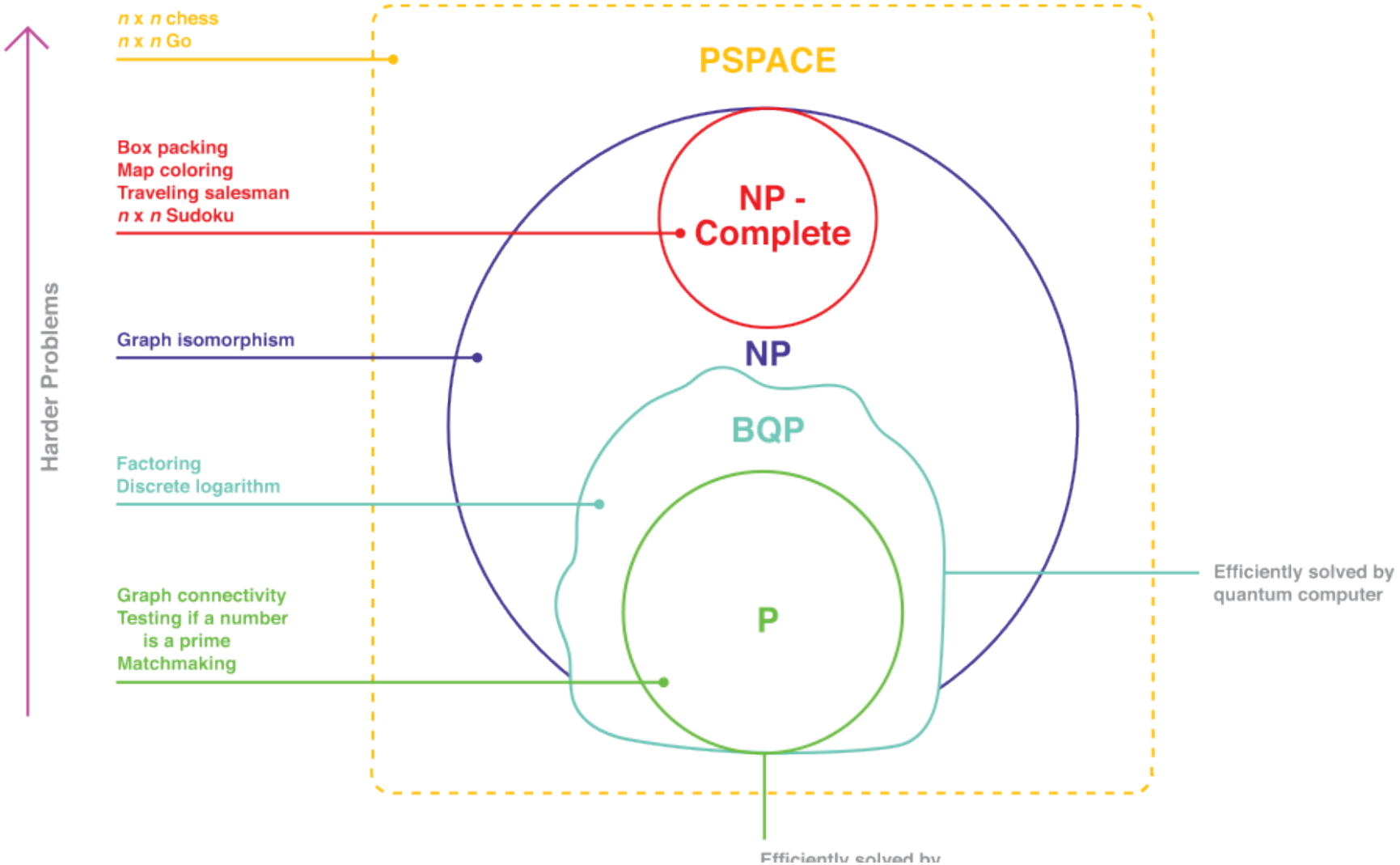
- Symmetric: exhaustive search, $\mathcal{O}(2^n)$
- Asymmetric: General Number Field Sieve, $\approx \mathcal{O}(2^{n-\epsilon})$

On Quantum Computers

- Symmetric: Grover's algorithm, $\mathcal{O}(\sqrt{2^n})$
- Asymmetric: Shor's algorithm, $\approx \mathcal{O}((\log n)^2)$

Shor's algorithm can break asymmetric cryptosystems in polynomial time!

The Computational Complexity Zoo



NIST Post-Quantum Cryptography Standardization

Timeline

- 26/02/16: Announcement of NIST's call for submissions
- 21/12/17: Announcement of the first round candidates (69)
- 30/01/19: Announcement of the second round candidates (26)
- 22/07/20: Announcement of the third round candidates (7 finalists, 8 alternatives)
- 05/07/22: Announcement of the fourth round candidates (4)
- 2022-2024: Publication of the standard drafts

Standardization candidates

Third Round winner

- CRYSTALS-Kyber: lattice-based

NIST decided to standardize at least another PQC scheme, in case lattice-based crypto fails

Fourth Round candidates

- ~~**S**upersingular **I**sogeny **K**ey **E**ncapsulation~~
- **B**it Fl**i**pping **K**ey **E**ncapsulation [ABB+22]
- **H**amming **Q**uasi-**C**yclic [MAB+18]
- Classic McEliece [BCC+22]

BIKE, Classic McEliece and HQC are all code-based cryptosystems.

Overview

The target of this thesis is Hamming Quasi-Cyclic

Goals

- High performance implementation for Cortex-M4
- Protection against timing attacks [GHJ+22, WBB+22]
- Protection against power attacks [SRSW20, SHR+22]

Roadmap

- Evaluate various polynomial multiplication algorithms
- Assess the constant-time behavior of all components
- Implement a masking scheme for the PKE primitives
- Compare the performance with the other Round 4 candidates

Background

Polynomial multiplication 101

$$a(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$
$$b(x) = b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n$$

Our goal: calculate:

$$c(x) = a(x) \cdot b(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_{2n} x^{2n}$$

- Schoolbook multiplication
- Karatsuba algorithm
- FFT-based approach

Schoolbook multiplication

Multiply each coefficient of $a(x)$ with each coefficient of $b(x)$

$$\begin{aligned} c(x) =& a(x) \cdot b(x) \\ =& a_0 \cdot (b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n) + \\ & + a_1 x \cdot (b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n) + power \\ & + \cdots + \\ & + a_n x^n \cdot (b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n) \end{aligned}$$

- n^2 multiplications
- n^2 additions

Overall complexity: $\mathcal{O}(n^2)$

Karatsuba multiplication

Recursively split $a(x)$ and $b(x)$ into two halves

$$a(x) = a_0(x) + x^m a_1(x) \qquad b(x) = b_0(x) + x^m b_1(x)$$

$$\begin{aligned} c(x) &= a(x)b(x) \\ &= (a_0(x) + x^m a_1(x))(b_0(x) + x^m b_1(x)) \\ &= a_1(x)b_1(x)x^{2m} + (a_1(x)b_0(x) + a_0(x)b_1(x))x^m + a_0(x)b_0(x) \\ &= z_2(x)x^{2m} + z_1(x)x^m + z_0(x) \end{aligned}$$

Trick: rewrite $z_1(x)$ as $(a_0(x) + a_1(x))(b_0(x) + b_1(x)) - z_0(x) - z_2(x)$

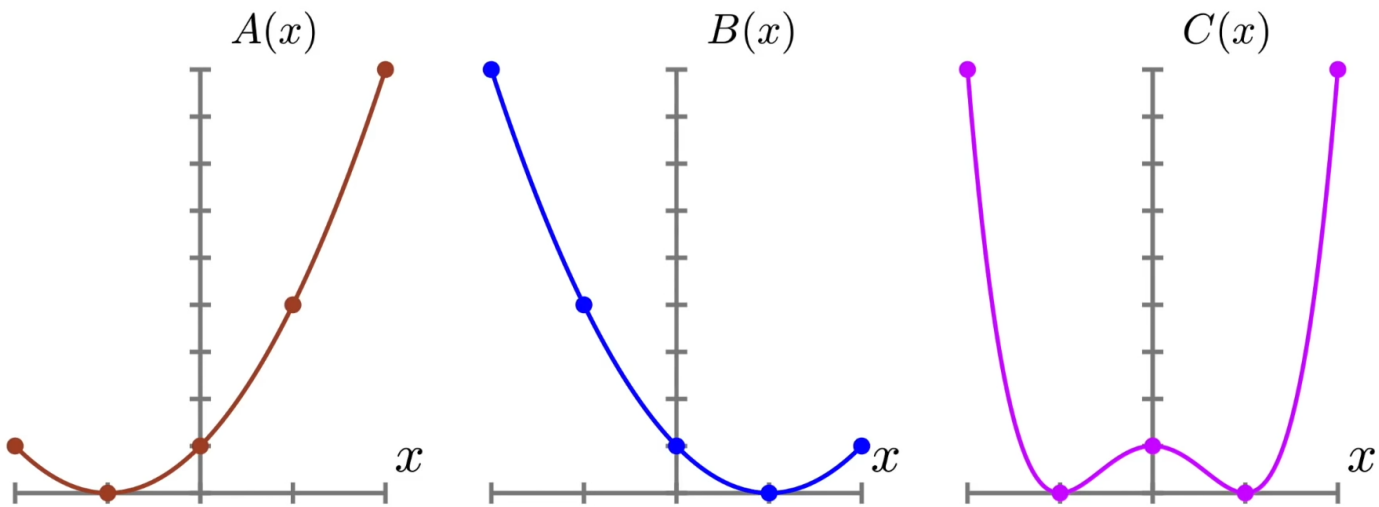
Overall complexity: $\mathcal{O}(n^{\log_2 3})$ if $m \approx n/2$

Sample and interpolate

- Sample $a(x)$ and $b(x)$ at $2n+1$ points
- Multiply each pair of samples
- Interpolate the points to get $c(x)$

$$A(x) = x^2 + 2x + 1 \qquad B(x) = x^2 - 2x + 1 \qquad C(x) = x^4 - 2x^2 + 1$$

$$[(-2,1),(-1,0),(0,1),(1,4),(2,9)] \qquad [(-2,9),(-1,4),(0,1),(1,0),(2,1)] \qquad [(-2,9),(-1,0),(0,1),(1,0),(2,9)]$$



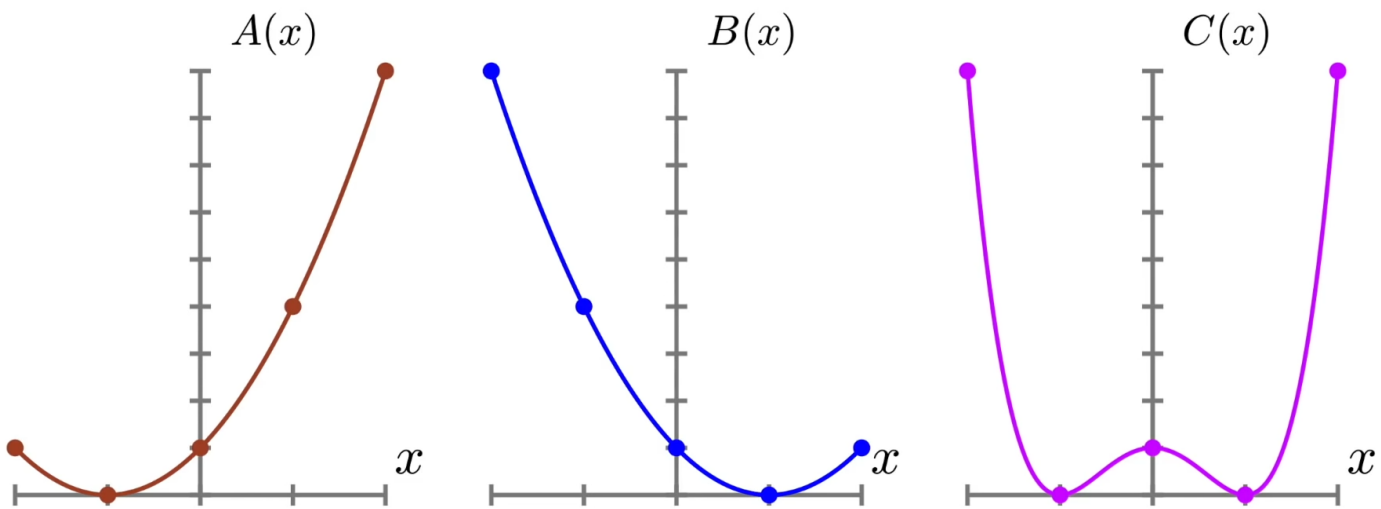
How can this be efficient?!

- Sampling: evaluate the polynomial at $2n+1$ points, $\mathcal{O}(n^2)$
- Pointwise multiplication: $\mathcal{O}(n)$
- Interpolation: up to $\mathcal{O}(n^3)$

$A(x) = x^2 + 2x + 1$ $[(-2,1),(-1,0),(0,1),(1,4),(2,9)]$

$B(x) = x^2 - 2x + 1$ $[(-2,9),(-1,4),(0,1),(1,0),(2,1)]$

$C(x) = x^4 - 2x^2 + 1$ $[(-2,9),(-1,0),(0,1),(1,0),(2,9)]$



The Discrete Fourier Transform (in $\mathbb{C}$)

Sampling

Evaluate the polynomial at the N-th roots of unity $\rightarrow$ take the DFT

$$\mathcal{F}(\omega_k) = \sum_{n=0}^{N} x_n e^{-\frac{2\pi i k}{N} n}$$

Interpolation

Interpolate the points $\rightarrow$ take the inverse DFT

$$\mathcal{F}^{-1}(X_k) = \frac{1}{N} \sum_{n=0}^{N} X_n e^{\frac{2\pi i k}{N} n}$$

Algorithms that calculate the DFT and its inverse are called Fast Fourier Transforms and can achieve $\mathcal{O}(n \log n)$ complexity.

FFT in finite fields

- $a(x)$ and $b(x)$ are polynomials in $\mathbb{F}_2[x]$
- Each coefficient $a_i, b_i \in \mathbb{F}_2$ is a bit (0, 1)

Problem: Finite fields (of characteristic 2) do not admit roots of unity
Solution: use powers of a primitive element
Caveat(s):

- The degree must be a power of 2: $n = 2^\ell$
- Polynomials require a different representation

Case study: the Lin-Chung-Han FFT [LCH14]

Cantor basis

Vector spaces and bases

$\mathbb{F}_2^m$: vector space of m-bit strings over $\mathbb{F}_2$, with m power of 2
Elements of $\mathbb{F}_2^m$ are *vectors*, linear combinations of the *basis vectors*
Basis vectors: $\mathbf{v} = (v_1, \cdots, v_m)$

Canonical basis

$$\mathbf{v} = (1, \alpha, \alpha^2, \cdots, \alpha^{m-1})$$

Cantor basis

$$\mathbf{v} = (v_1, \cdots, v_m)$$

$$v_1 = 1 \qquad v_i^2 + v_i = v_{i-1}$$

Subspaces and vanishing polynomials

$$v_1 = 1 \qquad v_i^2 + v_i = v_{i-1}$$

Subspaces induced by Cantor basis

$$\mathbf{W}_0 := \{0\}, \qquad \mathbf{W}_i := span\{v_0, \cdots, v_{i-1}\}$$

with $\mathbf{W}_0 \subset \mathbf{W}_1 \subset \cdots \subset \mathbf{W}_m = \mathbb{F}_2^m$ and $\dim(\mathbf{W}_i) = i$.

Vanishing polynomials

$$s_i(x) = \prod_{a \in \mathbf{W}_i} (x - a)$$

with $\deg(s_i(x)) = |\mathbf{W}_i| = 2^i$.

Novelpoly basis

Properties of the vanishing polynomials

- linearity: $s_i(x) + s_i(y) = s_i(x + y)$
- recursivity: $s_{i+j}(x) = s_i(s_j(x))$, $s_i(v_{i+j}) = v_j$
- $s_i(v_i) = 1$ for v_i element of the Cantor basis ([CCK+17])
- all v_i are roots of $s_{i+1}(x)$

Novelpoly basis

Cantor basis $\rightarrow$ Subspaces $\rightarrow$ Vanishing polynomials $\rightarrow$ Novelpoly basis

$$X_k(x) = \prod (s_i(x))^{b_i} \qquad \text{with } k = \sum_{i=0}^{\log_2 n} b_i 2^i$$

with $\deg(X_k(x)) = k$.

Polynomial evaluation (sampling)

We start from:

- $f(x) = g(X)$, a polynomial of degree 2^m in novelpoly basis
- $\mathbf{W}_m = \mathbb{F}_{2^m}$, the set of points to be evaluated

We want to get:

- $\{f(x) | x \in \mathbf{W}_m\}$, the evaluation of $f(x)$ at the points in $\mathbf{W}_m$

The LCH FFT

Divide et impera

Recursively split:

- The polynomial: $f_0(X) + f_1(X)X_{2^k}$
- The set of points: $\mathbf{W}_{k-1}, (\mathbf{W}_k \setminus \mathbf{W}_{k-1}) = \{w + v_{k-1} | w \in \mathbf{W}_{k-1}\}$

Get 2 for the price of 1

Evaluate f at $\mathbf{W}_{k-1}$; recall that:

- Elements in novelpoly basis are combinations of vanishing polynomials
- Vanishing polynomials are linear: $s_i(x) + s_i(y) = s_i(x + y)$

Thus the evaluations at $(\mathbf{W}_k \setminus \mathbf{W}_{k-1})$ can be obtained starting from the evaluations at $\mathbf{W}_{k-1}$ with a *butterfly* unit

LCH FFT - example

Consider a polynomial f in novelpoly basis of degree greater or equal than $\deg(s_k(x)) = 2^k$ and split it in half:

$$f(x) = p_0(x) + s_k(x)p_1(x)$$

Evaluate $p_0(x)$ and $p_1(x)$ at $\mathbf{W}_{k-1} = \{\alpha \mid \alpha \in \mathbf{W}_{k-1}\}$:

$$p_0(\alpha) + s_k(\alpha)p_1(\alpha)$$

Evaluate $p_0(x)$ and $p_1(x)$ at $(\mathbf{W}_k \setminus \mathbf{W}_{k-1}) = \{\alpha + v_k \mid \alpha \in \mathbf{W}_{k-1}\}$

$$p_0(\alpha + v_k) + s_k(\alpha + v_k)p_1(\alpha + v_k)$$

$p_0(x)$ and $p_1(x)$ are combinations of vanishing polynomials $s_i(x)$ with $i < k$:

$$\sum s_i(\alpha + v_k) = \sum s_i(\alpha) + s_i(v_k) = \sum s_i(\alpha) + 0 = \sum s_i(\alpha)$$

LCH FFT - example

From this consideration, we can observe that:

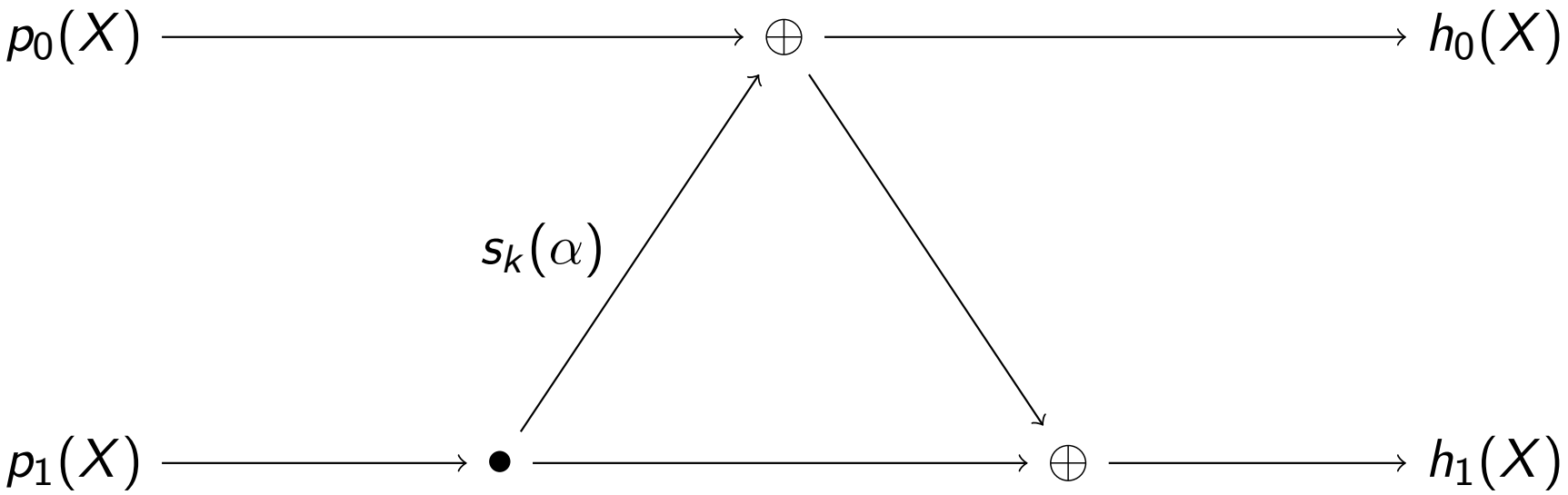
$$p_0(\alpha + v_k) = p_0(\alpha), \qquad p_1(\alpha + v_k) = p_1(\alpha)$$

Hence we can rewrite the evaluation at $(\mathbf{W}_k \setminus \mathbf{W}_{k-1})$ as:

$$p_0(\alpha + v_k) + s_k(\alpha + v_k)p_1(\alpha + v_k) = p_0(\alpha) + s_k(\alpha)p_1(\alpha) + s_k(v_k)p_1(\alpha)$$

and thus reuse the first evaluation to get the second one.

Figure: Butterfly unit



Frobenius map

We can use the Frobenius map

$$\phi : \mathbb{F}_2 \to \mathbb{F}_2 \qquad x \mapsto x^2$$

to derive more values of a polynomial f of degree n from less evaluations.

Frobenius partition

We can build a set of points $\mathbf{S} \subset \mathbb{F}_{2^m}$ as:

$$\mathbf{S} := v_{l+m/2} + \mathbf{W}_l = \{v_{l+m/2} + x \mid x \in \mathbf{W}_l\}$$

which has size $n' = |\mathbf{S}| = |\mathbf{W}_l| = n/m$. By repeatedly applying ϕ to $\mathbf{S}$ we get

$$\mathbf{S} \cup \phi(\mathbf{S}) \cup \cdots \cup \phi^{m-1}(\mathbf{S}) = \mathbf{O}$$

with $|\mathbf{O}| = n$.

Implementation

Experimental results

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